

DBMS Lab

SQL Experiment-2

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Section: AF1

Roll no: 22

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- (i) Add cName as Primary key in College.

```
SQL> alter table college
      add primary key(cName)

Table COLLEGE altered.

Elapsed: 00:00:00.034
```

- (ii) Add sID as Primary key in Student.

```
SQL> alter table student
      add primary key(sID)

Table STUDENT altered.

Elapsed: 00:00:00.220
```

- (iii) Add sID, cName, major as Primarykey in Apply.

```
SQL> alter table apply
      add primary key(sID,cName,major)

Table APPLY altered.

Elapsed: 00:00:00.034
```

- (iv) Make sID in Apply foreign key referring table student and cName referring table college.

```
SQL> alter table apply
      add constraint fk_apply_student
      foreign key(sID)
      REFERENCES student(sID)

Table APPLY altered.

Elapsed: 00:00:00.021
```

```
SQL> alter table apply
      add constraint fk_apply_college
      foreign key(cName)
      REFERENCES college(cName)

Table APPLY altered.

Elapsed: 00:00:00.032
```

- (v) Increase data type size of major from 20 to 25.

```
SQL> alter table apply
      modify(major varchar2(25))

Table APPLY altered.

Elapsed: 00:00:00.023
```

- (vi) Add a new column decision in the Apply table keeping a constraint of not null for this column with data type varchar2(3).

```
SQL> alter table apply
      add(decision varchar2(3) not null)

Table APPLY altered.

Elapsed: 00:00:00.033
```

- (vii) Change data type of decision in Apply to char(1).

```
SQL> alter table apply
      modify(decision char(1))

Table APPLY altered.

Elapsed: 00:00:00.023
```

- (viii) Drop foreign key on column name cName from Apply table.

```
SQL> alter table apply
      drop constraint fk_apply_college

Table APPLY altered.

Elapsed: 00:00:00.024
```

- (ix) Remove column sizeHS from Student table.

```
SQL> alter table student  
      drop column sizeHS
```

```
Table STUDENT altered.
```

```
Elapsed: 00:00:00.037
```

(x) Drop primary key from College

```
SQL> alter table college  
      drop primary key
```

```
Table COLLEGE altered.
```

```
Elapsed: 00:00:00.033
```

(xi) Make cName, major unique pair wise such as Stanford CS, Stanford EE.

```
SQL> alter table apply  
      add constraint uq_cname_major  
      unique (cName, major)
```

```
Table APPLY altered.
```

```
Elapsed: 00:00:00.024
```

(xii) Add cName as Foreign Key in Apply table referring table College using on delete cascade.

```
SQL> alter table college  
      add constraint uq_college_cname unique (cName)
```

```
Table COLLEGE altered.
```

```
Elapsed: 00:00:00.028
```

```
SQL> alter table apply
      add constraint fk_apply_college
      foreign key (cName)
      references college(cName)...
Show more...

Table APPLY altered.

Elapsed: 00:00:00.031
```

- (xiii) Modify foreign key on sID in Apply table to foreign key on delete set null.

```
SQL> alter table apply
      drop constraint fk_apply_student

Table APPLY altered.

Elapsed: 00:00:00.026
```

```
SQL> ALTER TABLE Apply
      ADD CONSTRAINT fk_apply_student
      FOREIGN KEY (sID)
      REFERENCES Student(sID)...
Show more...

Table APPLY altered.

Elapsed: 00:00:00.024
```

- (xiv) Rename column enrollment to enroll in College Table.

```
SQL> ALTER TABLE College  
      RENAME COLUMN enrollment TO enroll
```

```
Table COLLEGE altered.
```

```
Elapsed: 00:00:00.017
```